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Fertilizer Recommendation System Using Machine Learning Classifiers

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ABSTRACT

Selecting the right fertilizer for a given crop and soil condition is a critical yet often inefficient process for farmers, who traditionally rely on generic guidelines or trial-and-error methods that can lead to poor yield, wasted resources, and soil degradation. This project presents a Fertilizer Recommendation System that uses supervised machine learning classifiers to predict the most suitable fertilizer based on key parameters such as soil type, crop type, temperature, humidity, moisture, and nutrient levels (nitrogen, phosphorus, and potassium). The dataset is first preprocessed through cleaning, encoding of categorical features, and scaling of numerical attributes to ensure the models receive well-structured input. Multiple classification algorithms, including Decision Trees, Random Forest, and Logistic Regression, are trained and compared to identify the model that delivers the highest prediction accuracy, evaluated using metrics such as accuracy, precision, recall, and F1-score. The best-performing model is then used to build a simple recommendation interface where a user can input soil and crop details and instantly receive a fertilizer suggestion. By automating this decision-making process, the system aims to help farmers make informed, data-driven choices, improve crop productivity, and promote sustainable and cost-effective use of fertilizers.

THANK YOU